

KAIROSRANK: A PROACTIVE, JUST-IN-TIME AGENT THAT DECIDES WHEN TO BREAK SILENCE, WHAT TO ASK, AND HOW TO RE-RANK

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ABSTRACT

Most sites already personalize their listings, but they do so *passively* and server-side: the ordering is recomputed only when the user reloads the page or feeds the system more explicit input (a new query, a filter, a click), and it is optimized against the platform’s objective rather than the user’s live intent. What is missing is a *proactive, within-visit* adjustment that reads the user’s behavior as it happens and re-orders in place, on the user’s side, without a refresh or an explicit request. We present **Kairosrank**, a browser extension that acts as a *proactive*, just-in-time information agent: without ever being asked, it senses the user’s journey (scrolls, hovers, dwell, searches), infers on its own initiative when help is worth the interruption, and *silently re-orders the existing list in place* to match a learned taste profile—without a page re-render and without an explicit request. Being proactive is the whole point: the system takes the first move, deciding on every tick whether acting is worth more than staying silent, rather than waiting for the user to type, filter, or click. The contribution is twofold. As a *research contribution*, we take the just-in-time information recommendation (JIR) problem of Yang et al. (2025b)—which they cast as a POMDP—and turn it into a concrete, deployable interaction design: rather than solving a POMDP, we approximate its act-only-when-worth-it objective with cheap on-device estimators and hard interaction structure, governed by six non-negotiable requirements—user-as-principal, precision-over-recall, reversibility, earned escalation, local-first, and page integrity—and we decompose the agent into a two-tier decision architecture in which *when to act* is a separate, learned module from *what to deliver*. As an *engineering contribution*, we build this architecture under the hard constraints of Manifest V3: a 384-dimensional multilingual sentence embedder runs fully on-device in a WASM offscreen document; a per-site preference centroid drives an instant cosine re-rank; and a 16-feature online logistic “when-to-interact” classifier decides each tick whether to fire, learning from the user’s reactions via stochastic gradient descent. A cloud large language model is reserved for the one case only it can serve—an opinionated clarifying question when the profile is thin. The prototype spans five sites (one video, four research/ML-discovery), is validated by 61 unit tests across 13 suites and by end-to-end Playwright runs against the unpacked extension, and re-ranks a detected list on-device in milliseconds. We report the design, the implementation, its honest limitations, and a phased plan for the user-facing evaluation the system is instrumented to produce.

1 INTRODUCTION

The dominant failure mode of the modern web is not missing information but mis-ordered information. A user arrives at a listing—search results, an arXiv “recent” page, a model hub, a video grid—carrying a specific, transient intent. Modern listings are rarely un-personalized; large platforms rank with sophisticated per-user models. But that personalization is *passive* and *server-side*:

*Research & development report. The system described is an open engineering prototype; quantitative user-facing evaluation is scoped as future work (Section 6).

it is computed from long-run history and platform objectives, delivered at page-load, and updated only when the user refreshes or supplies more explicit input—a new query, a filter toggle, another click. It does not read what the user is doing *during this visit*, and it cannot re-order the list in front of them without a round-trip. So a user whose intent shifts mid-session—who realizes three items in that they want theory papers, not applications—has no recourse but to act on the system’s terms: re-query, re-filter, scroll. The cost is paid in scanning, scrolling, `Ctrl-F`, and pogo-sticking back to the search box. Our premise is that the missing capability is not better server-side ranking but a *proactive, local, within-visit* agent that infers the shift from behavior and re-orders in place—no refresh, no explicit request, no network round-trip. We scope this deliberately to the *within-page* problem: the agent re-ranks the items the site has already put on the page, rather than retrieving new ones from across the site—improving the ordering of a given candidate set, not enlarging it.

Yang et al. (2025b) formalize the general version of this problem as *just-in-time information recommendation* (JIR): surface the right information at the right moment without being asked, by inferring a user’s latent need from ongoing context and delivering within an effective time window. They cast it as a partially observable Markov decision process (POMDP) and evaluate systems on three axes—inferring needs, timely relevant delivery, and avoiding distraction—finding that foundation-model pipelines reach reasonable *precision* but weak *recall*. That finding is the design premise of this work: if precision is achievable but recall is not, an agent that acts should prefer silence to a wrong intervention, and treat every visible action as a cost to be justified.

We present **Kairosrank**, a browser extension that turns this premise into a running system on a deliberately narrow but real slice of the JIR problem: re-ordering the items already on a page to fit what the user wants *right now*. The system is deployable today, runs its learning and inference entirely on the user’s device, and treats a cloud call as a rare, consented escalation rather than the default path.

We separate our contribution into two components, following the structure a research-and-development report is expected to have.

Research component (Section 3). We take the JIR objective as our target and turn it into an interaction design, without claiming to solve the underlying POMDP: our belief and policy are heuristic estimators, not a maintained posterior or a planned policy. Three ideas do the work. (i) A set of six *non-negotiable requirements* that convert the “precision over recall” finding into hard architectural constraints—most importantly that silence is the default action and that any page modification is reversible in one interaction. (ii) An *earned-escalation action ladder*: the agent may only climb from ambient hints to content transformation to a dialogic question as per-site trust accrues, and a single undo re-locks the rung. (iii) A *two-tier decomposition* in which the decision of *when* to interrupt is a distinct, independently learned module from *what* to deliver—mirroring, and motivated by, the separation between interruptibility modeling (Mehrotra & Musolesi, 2017; Zheng et al., 2025) and need inference in the literature.

Engineering component (Sections 4–5). We show that this design is buildable under Manifest V3 (Google Chrome Developers, 2024), whose security model forbids remotely loaded code and blocks dynamic imports in the service worker. Our system runs a quantized 384-dimensional multilingual sentence embedder (Reimers & Gurevych, 2019) on-device in an offscreen document via WASM (Hugging Face, 2024); maintains a per-site preference centroid updated by engagement signals; ranks list items by cosine similarity and re-orders the live DOM non-destructively; and gates every proactive action behind a 16-feature online logistic classifier that learns per site from the user’s reactions. The cloud LLM is invoked only to generate a single grounded clarifying question when local confidence is insufficient, with speculative scoring hiding its latency.

The prototype covers five sites, is verified by 61 unit tests across 13 suites and by Playwright end-to-end runs loading the actual unpacked extension, and performs a silent on-device re-rank in milliseconds. We are candid (Section 6) that the system does not yet report user-facing metrics on our three evaluation dimensions (act-value, timing, restraint); it is, however, instrumented precisely to produce them, and we lay out the phased plan for doing so.

2 BACKGROUND AND RELATED WORK

Our design sits at the meeting point of two literatures that have largely evolved apart: *just-in-time information retrieval*, which removes the explicit query and triggers retrieval from ambient context—from the Remembrance Agent (Rhodes & Starner, 1996) and the JITIR formulation of Rhodes & Maes (2000) to the POMDP synthesis of Yang et al. (2025b)—and a recent line of *proactive agents* that equip models to decide when to retrieve, act, or ask. Rather than survey these in the aggregate here, we defer the detailed treatment to the research component (Section 3): because we argue each research challenge is forced by a product requirement, the related work most usefully appears *beside the challenge it bears on*, where we can also say precisely what each prior result leaves unsolved for an on-device, in-browser, reversible re-ranking agent. This section states only the shared frame; each of Sections 3.1–3.5 then carries its own related work and residual gap.

The shared operating point. What unifies our reading of both literatures is a single distinguishing choice of operating point: *local-first, precision-first, and reversible*. The field’s timing signals, preference models, idle-time pre-compute, and usefulness-gated clarification all reappear in our system, but subordinated to a browser-deployable design whose default action is to do nothing and whose most aggressive action a user can undo with one click. Because each of these threads bears on a different one of our design challenges, we pair each with the specific research question it informs: Sections 3.1–3.5 each open with the related work and the precise residual gap for that question, rather than treating the literature as one undifferentiated block.

3 RESEARCH COMPONENT: RESEARCH QUESTIONS DERIVED FROM PRODUCT REQUIREMENTS

The research questions in this section are not chosen for their theoretical interest; each is *forced* by a requirement of the product we set out to build—a browser agent that re-ranks pages for the user’s live need without ever costing them. We therefore state each question together with the requirement that generates it, so that “is this research question worth answering?” reduces to “does the product need it answered?” The requirements and metric targets we cite (NR-1–6, FR-1–8) are those of the system’s requirements specification, which was seeded by the JIR-Arena problem and the project’s literature survey; the evaluation itself, however, we frame on our own dimensions (act-value, timing, restraint), read off the decision rule rather than borrowed.

Motivation and framing. Yang et al. (2025b) cast just-in-time recommendation as a POMDP, and we borrow that vocabulary to *state* the problem. In the frame, a hidden state is the user’s latent need n_t and the window in which it is live; the agent never observes it, seeing only o_t —the DOM structure plus a behavioral event stream (scrolls, hovers, dwell, opens, searches), the same GUI-interaction modality PIRA-Bench (Chai et al., 2026) targets—and at each tick choosing an action a_t from a ladder whose default element is **no-op**, for which it receives only a delayed, censored proxy of whether it helped. We borrow two things from Yang et al. (2025b): the *problem definition*, and—most consequentially—the *empirical finding* that such systems reach usable precision but weak recall, which fixes our operating point. We do *not* implement the POMDP machinery: no learned transition model, no maintained Bayesian posterior over n_t , no planning or policy-solving. Where the frame would keep a belief state we substitute a cheap online estimate; where it would solve a policy we substitute a hand-structured action ladder plus an online-learned binary gate. Equation 1 is thus an expected-value *decision rule* we approximate on-device, not a POMDP solution—the gap is deliberate, since the approximations are what fit a content script’s compute and latency budget. Nor do we adopt their three evaluation axes as our own: JIR-Arena scores systems against a curated ground truth of “true needs,” but our system has no such oracle at run time, so we instead evaluate against dimensions read directly off this decision rule, defined next.

The product requirements, and why each is needed. Because we claim each research question is *forced* by the product, we first state the product’s requirements and justify why each one is not optional. The system is bound by six non-negotiable requirements—NR-1 *user is principal* (no sponsored or engagement objective), NR-2 *precision over recall / silence by default*, NR-3 *reversibility*, NR-4 *escalation is earned*, NR-5 *local-first*, NR-6 *page integrity*—each of which exists

FR	What the product must do	Why it is needed
FR-1	Sense page structure and behavior locally at $\leq 10\text{Hz}$; never capture keystrokes	Supplies the only observation stream o_t the agent has; must be local to satisfy NR-5, and keystroke-free to be privacy-defensible.
FR-2	Two-stage need inference: a $< 50\text{ms}$ local stage-1, an LLM stage-2 only when warranted	Scoring need on every tick with an LLM is infeasible on latency and cost (Section 4.6); the cheap local stage is what makes continuous sensing affordable.
FR-3	A timing gate, <i>separate</i> from need scoring, firing only on a breakpoint within budget	<i>When</i> to interrupt is a different problem from <i>what</i> to say (PersoNo); coupling them causes mistimed, distraction-inducing actions [T,R].
FR-4	A four-rung action ladder (ambient / additive / transform / dialogic) with per-rung dismiss and provenance	One fixed action cannot match a varying confidence and cost; a graduated ladder bounds the worst-case intrusion and makes escalation auditable.
FR-5	Questions as one-tap chips (2–4 options), behind a usefulness gate, with a visible payoff	Some needs cannot be served by re-ordering existing items; the question is the costliest rung, so it must be cheap to answer, worth asking, and immediately useful or it is pure interruption.
FR-6	Per-site memory of accept/undo/ignore, with thresholds that adapt and rungs that lock on undo	Taste and tolerance are per-user and per-site and must persist; this store is the mechanism that makes escalation <i>earned</i> (NR-4) rather than default.
FR-7	Global pause, per-site off, an activity log, and always-available restore	An agent that modifies pages is trustable only if the user can override and audit it—the operational form of NR-1 and NR-3.
FR-8	Idle-time precompute of likely payloads, under a hard per-page LLM-call budget	Delivery must be instant at trigger time [T] even though LLM calls are slow and costly; precompute within a fixed budget reconciles the two.

Table 1: The eight functional requirements of the product, each with the justification for why it is required. Requirement text is condensed from the system’s requirements specification; these IDs are the “forcing requirements” referenced in Table 2.

because its violation is a distinct way to lose the user: a hidden agenda, a noisy agent, an irreversible mistake, an over-eager one, a privacy breach, or a broken page. On top of these, eight functional requirements (Table 1) specify what the system must *do*; each is justified by a failure it prevents or a capability the objective of Equation 1 cannot be met without.

Five research questions. Read at the level a product must be built to, meeting these requirements decomposes into five coupled research questions. RQ1 asks what the objective even is; RQ2 and RQ3 are the classic *when/what* split; and because “what to deliver” silently conflates *re-ordering items* with *asking the user something*—two acts with different failure modes—we separate the question act into RQ4 (*whether* to ask) and RQ5 (*how* to ask), the latter a design question the prototype only begins to answer. Table 2 maps each question to the requirement above that forces it.

Our evaluation dimensions. Rather than inherit JIR-Arena’s need/timeliness/distraction axes—which presuppose a ground-truth “true need” our run-time system never sees—we read three evaluation dimensions directly off our own decision rule, the act-only-when-worth-it gate of Equation 1. **[V] Act-value:** when the agent does act, is the act worth it? This is the realized value of the $\mathbb{E}[\text{value} \mid \text{act}]$ term, measured behaviorally (engagement with re-ranked items, question answer-and-change rate). **[T] Timing:** given that an act is worthwhile, does it land at a low-cost moment? This is the moment-dependence of the $\text{cost}(a_t, t)$ term (interaction lag, and the rate at which a worthwhile act expires unseen). **[R] Restraint:** is silence correctly the default? This measures the no-op default and the reversibility that bounds a wrong act (undo rate, ignore rate, per-site disables, and the “canary” rate of visible actions during moments that should have stayed quiet). The three are not the JIR axes relabeled: they are the three terms one can point to in Equation 1, and each RQ below is evaluated against the one(s) its decision most directly moves.

RQ	Research question	Forcing product requirement	Evaluation (axis)
RQ1	What does it mean to be <i>right</i> ? (the objective)	NR-1, NR-2 (silence default); goal “decisive, never annoying” (R-3)	M0 stage-1 precision; canary set [R]
RQ2	<i>When</i> to break silence?	FR-3 timing gate; FR-3.3 budgets; NFR-1 latency	lag, unseen-expiry [T]
RQ3	<i>What</i> ordering to deliver?	FR-2 need inference; FR-6 memory	engagement, usefulness [V]
RQ4	<i>Whether</i> to ask a question?	FR-5.2 usefulness gate; open question Q-6	gate hold-rate, ask→change [V,R]
RQ5	<i>How</i> to ask it? (form/rationale)	FR-5.1 (2–4 options); FR-4.3 provenance	answer rate, dismissal [V,R]

Table 2: The five research questions of the research component, each traced to the product requirement that forces it and to the evaluation dimension (and metric) that answers it. The dimensions [V] act-value, [T] timing, and [R] restraint are our own, read off the decision rule of Equation 1 (defined above), not JIR-Arena’s axes. Requirement IDs are those of the system’s requirements specification; the evaluation plan is detailed per-RQ below and consolidated in Section 6.

Each subsection below states the question and its forcing requirement, formalizes the challenge as an equation, argues why it is genuinely hard, places it against the related work, gives our current solution (distinguishing what is *built and tested* from what is *designed and instrumented*), and states how we will evaluate it. A deliberate through-line is that each per-challenge equation is a *special case of a more general object the cited literature already studies*: the general form is shared with prior work, and our contribution is the specialization—a myopic reduction (RQ1), an online censored-feedback estimator (RQ2), a cosine-against-centroid score (RQ3), a re-ranking-valued information gate (RQ4)—forced by the on-device, reversible, in-browser setting. RQ5 is the honest exception, where our objective *extends* rather than specializes the literature’s by one coupling term; we flag it as such. A recurring theme is that the questions are easier together than apart: the objective’s asymmetry (RQ1) is what licenses a cheap, aggressive timing model (RQ2), because the delivery side (RQ3) guarantees every act is reversible.

3.1 RQ1 — THE PRIMARY OBJECTIVE: WHAT DOES IT MEAN TO BE RIGHT?

Question and forcing requirement. Before timing or content, the product needs a definition of success that its own metrics can be built on. The requirement is explicit and unusual: NR-1 makes the browsing user the sole principal (no engagement or sponsored objective), NR-2 makes *silence the default action* and precision the priority over recall, and the stated product goal is to be “occasionally decisive, never annoying” (risk R-3). So the research question is: formalize an objective under which a missed need is cheap but a wrong interruption is expensive, and which a shippable metric can track. A conventional recommender maximizes engagement or ranking accuracy over a fixed labeled set; this product may not. We state the objective as an expected-value gate—act only when

$$\underbrace{\hat{P}(n_t \mid o_{1:t})}_{\text{estimated need score}} \times \underbrace{\text{value}(a_t \mid n_t)}_{\text{help if right}} - \underbrace{\text{cost}(a_t, t)}_{\text{interruption + risk of wrong act}} > 0. \quad (1)$$

The research problem is to make every term of Equation 1 estimable and its trade-off honest on a real device.

Why it is hard. Three properties defeat a naive treatment. (i) *The loss is asymmetric and trust-coupled.* A single wrong interruption does not cost one unit of accuracy; it can cost the agent its standing permission to touch the page at all, whereas a missed need costs almost nothing—the user simply keeps browsing as they would have. No symmetric metric (accuracy, F1) encodes this, which is exactly why Yang et al. (2025b) report that foundation-model pipelines reach usable precision but weak recall, and why optimizing recall would be the wrong target. (ii) *The reward is censored and confounded.* The agent observes an outcome only for ticks on which it acted—staying silent yields no counterfactual—and even the observed signal (an open within the window) is a noisy, delayed proxy that the re-rank may not have caused. This is a contextual-bandit reward, not a supervised label. (iii) *cost(a_t, t) is not a constant.* The same re-rank is nearly free during an idle scan and expensive mid-read; cost depends on the action’s own intrusiveness *and* on the moment.

Related work and the residual gap. The objective inherits from two sources that each supply half of it and neither completes. Classical JITIR (Rhodes & Starner, 1996; Rhodes & Maes, 2000) established that an agent may help without being asked and must do so non-intrusively, and justified low-cost peripheral presentation—but it argued cost from HCI intuition, never as an *asymmetric-loss decision rule*, and its agents were strictly additive (a margin note), so they never confronted the trust-coupling that arises once the agent modifies the primary content. Yang et al. (2025b) supply the missing decision theory: they cast the problem as a POMDP, define the three axes (need inference, timeliness, distraction), and—the result we lean on hardest—measure that foundation-model pipelines reach usable precision but weak recall. In the terms of Eq. 1, their POMDP is the general object and our gate is its *myopic* (one-step, greedy) specialization—we maximize the immediate expected value at each tick rather than a discounted return—which is precisely the reduction that makes the decision computable on every event on-device. Rhodes, by contrast, offers no formal object to specialize: the JITIR account is a qualitative precursor, so the “general form” relationship holds for JIR-Arena but not for it. But JIR-Arena is a *benchmark and synthesis*, not a deployable operating objective: it scores systems on the axes without prescribing the hard non-negotiable requirements a system must obey to keep the loss asymmetric under a live user, and it does not treat reversibility as a term that changes the cost. Staged-retrieval-with-feedback (Sloan & Wang, 2016) optimizes retrieval quality across rounds, again with symmetric relevance objectives rather than an act/no-act gate whose false positives cost standing permission. The residual gap our RQ1 occupies is therefore the step none of them takes: turning the measured precision-over-recall asymmetry into a small set of *non-negotiable, machine-checkable requirements plus a cost ladder* that a shipped, page-modifying agent cannot optimize away—and making *reversibility itself* a cost-reducing term, which is what lets the gate be crossed on modest belief. What stays genuinely hard even after this framing is that the objective can never be directly validated: with a censored, confounded reward there is no ground-truth label for “was silence correct here,” so success can only be triangulated from the distraction-avoidance canaries staying quiet while the agent remains occasionally decisive.

Our current solution. We make the objective operational through structure rather than a single learned reward. First, six **non-negotiable requirements** hard-code the asymmetry so no learned component can trade it away: (1) the user is the sole principal (no sponsored or engagement objective); (2) precision over recall—silence is the default and a missed need is preferred to a wrong act; (3) reversibility—every page change is undoable in one interaction; (4) escalation is earned, never default; (5) local-first—raw content and telemetry stay on device unless a cloud call is consented; (6) page integrity—non-destructive DOM operations only, auto-reverting on any layout error. Requirement NR-3 (reversibility) is what makes acting-under-uncertainty affordable: because a wrong act is one click from undone, the realized cost of a false positive is bounded, so Equation 1 can be crossed on modest belief. Second, we make $\text{cost}(a_t, t)$ explicit and monotone with an **earned-escalation action ladder** (Figure 1): R1 an ambient anchored hint, R2 an additive margin card that never touches host layout, R3 the content *transformation*—the silent re-rank, offered with a preview and a persistent restore—and R4 a single one-tap clarifying question. Each rung is gated by a confidence threshold τ , and R3/R4 are additionally gated by *earned per-site tolerance*: the agent climbs only as accepted interactions accumulate on that site, and one undo re-locks the transform rung for a cooldown. This operationalizes the ramping-interface lesson of Rhodes & Maes (2000)—trust granted incrementally, revoked instantly—and turns the abstract cost term into a discrete, auditable ladder the user’s own history moves up and down.

Status & evaluation. The non-negotiable requirements and ladder are *built* (the ladder’s gates and undo-relock are unit-tested; Section 5.2); the objective itself is a design claim, not yet an empirical result. It is validated by the distraction-avoidance canary set: ignore rate, R3 undo rate ($< 10\%$ target), per-site disables, and global pauses, with the standing rule that any canary breach freezes all threshold loosening and rung unlocks. If the objective is correctly stated, these canaries stay quiet while the agent is still occasionally decisive—the two together, not either alone, are the test.

3.2 RQ2 — WHEN TO BREAK SILENCE?

Question and forcing requirement. FR-3 requires a timing gate that is a *separate module* from need scoring (the PersoNo pattern: when-to-interrupt is independent of what-to-deliver), FR-3.3 imposes hard interruption budgets, FR-3.4 forbids acting mid-scroll or mid-selection, and NFR-1

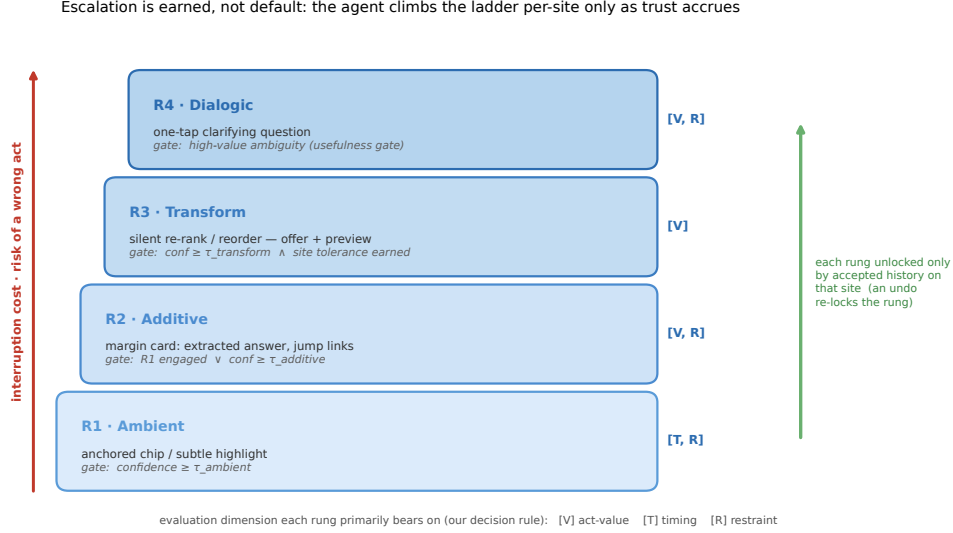


Figure 1: The earned-escalation action ladder, our operational model of the $\text{cost}(a_t, t)$ term in Equation 1. Rungs are ordered by interruption cost and risk of a wrong act (left axis). R1–R2 are gated by confidence alone; R3 (silent re-rank) and R4 (clarifying question) additionally require per-site tolerance earned from accepted history, and an undo re-locks the rung (right rail). Bracketed tags map each rung to the evaluation dimension it primarily bears on, in our own terms (Equation 1): [V] act-value, [T] timing, [R] restraint.

caps the decision at tens of milliseconds. The research question follows: at every tick, output a binary break-silence / stay-quiet choice from the running behavioral stream $o_{1:t}$, with no explicit query to anchor on—estimating $\hat{P}(n_t | o_{1:t})$ well enough, at the right moment, and cheaply enough, to decide whether the gate of Equation 1 is crossed.

Why it is hard. The difficulty is not classification accuracy in the abstract; it is the setting. (i) *Feedback is one-sided (censored bandit)*. The agent learns the outcome of a tick only if it acted there; the silent majority of ticks produce no label, so it cannot simply be trained as supervised interruptibility. (ii) *No population training set fits*. Taste and rhythm are per-user and per-site and non-stationary within a single session (a user hunting for one paper behaves nothing like the same user browsing videos an hour later), so the model must learn online from a cold start rather than ship pre-trained. (iii) *Positive events are rare and cheap signals are ambiguous*. Opens and saves are sparse relative to the thousands of scroll/hover events, and interruptibility is not the same as need—a user can have a strong need at a terrible moment to interrupt. (iv) *It must be nearly free*. The decision runs on every event, on-device, inside a content script, so the model has a microsecond-scale compute budget.

Formalization. At tick t the agent emits a binary $d_t \in \{0, 1\}$ (act / stay silent) from history $o_{1:t}$. Writing the estimated need score as a logistic model $\hat{P}(n_t | o_{1:t}) = \sigma(w^\top \phi(o_{1:t}))$ over the journey features $\phi \in [0, 1]^{16}$, the myopic form of the master gate (Eq. 1) reduces to a threshold rule

$$d_t = \mathbb{I}[\sigma(w^\top \phi(o_{1:t})) \geq \tau], \quad \tau = 0.6, \quad (2)$$

but the weights w must be learned from *censored bandit* feedback: a reward $r_t \in \{0, 1\}$ (an open within a window Δ) is observed only where $d_t = 1$, so the online update sees the loss

$$\ell_t(w) = -d_t [r_t \log \hat{p}_t + (1 - r_t) \log(1 - \hat{p}_t)], \quad \hat{p}_t = \sigma(w^\top \phi(o_{1:t})), \quad (3)$$

which is identically zero on every silent tick. The learner therefore optimizes over a support it selects by acting—the structural reason a too-cautious prior can starve its own gradient.

Related work and the residual gap. Timing has its own lineage, and every branch of it stops short of the constraints above. The interruptibility literature (Mehrotra & Musolesi, 2017), its large-scale field studies (Pielot et al., 2017; Okoshi et al., 2017), and its recent successor PersoNo (Zheng et al., 2025) establish the design principle we adopt—*when* to interrupt is a distinct problem from *what* to deliver—but they learn interruptibility from population data collected offline, whereas our setting has no population training set that transfers: rhythm and tolerance are per-user, per-site, and non-stationary within a session. Okoshi et al. (2017) are the closest operational ancestor: they detect *breakpoints* in activity and defer notifications to them, cutting response time at the scale of hundreds of thousands of users—but a breakpoint detector fired server-side against a notification queue is not a per-event, on-device gate deciding whether to silently re-order the page a user is looking at, and their signal is a fixed heuristic rather than an online per-site classifier. Sequence models that predict need from action histories (Luukkonen et al., 2016) demonstrate that behavioral streams carry the signal, but as offline-trained predictors, not online learners that start from a cold, opinionated prior and adapt from the user’s own reactions. The adaptive-retrieval family makes timing *internal to generation*—Self-RAG triggers retrieval from learned reflection tokens (Asai et al., 2023), and later methods time evidence injection by step-level uncertainty (Guo et al., 2026) or iterative gap detection (Aghajani Asl et al., 2025)—but “when to retrieve inside an LLM’s decoding loop” is a server-side decision with a GPU behind it, not a microsecond dot product on every scroll event in a content script. Context-anticipating agents such as ContextAgent (Yang et al., 2025a) come closest in spirit, inferring needs from continuous sensory streams, yet they assume rich multimodal sensing and server compute rather than a privacy-bounded on-device feature vector. Each of these is, in the terms of Eq. 2, the same general object—a decision $d = \mathbb{I}[g(\text{context}) \geq \tau]$ for some estimated need-or-interruptibility score g —and they differ only in how g is obtained: an offline population model (interruptibility), an offline sequence predictor (LSTM), or a decoding-time uncertainty (adaptive retrieval). Our formulation is the specialization in which $g = \sigma(w^\top \phi)$ is learned *online, per site, from censored feedback* (Eq. 3); the general threshold rule is shared, the estimation regime is what we change. The residual gap is the conjunction none of them meets at once: an interruptibility decision that is *online* (no pretraining set), *per-user and per-site* (non-stationary, cold-start), learned from *censored bandit feedback* (a label only where it acted), and cheap enough to run on *every event on-device*. What remains hard even given our design is the censoring itself—because silent ticks yield no counterfactual, the timing model can only ever learn from the moments it already chose to break silence, so a prior that starts too cautious can suppress the very evidence that would correct it, which is why the hand-set prior and the shadow-mode calibration phase are load-bearing rather than incidental.

Our current solution. A single **online logistic-regression classifier** over a 16-dimensional journey feature vector (Figure 3), learned *per site*. Features summarize the live session—scroll distance, hovers, opens, bounces, search/filter “hunts,” dwell, time-since-last-open, a no-open-yet flag, dissatisfaction, late-night, and six research/discovery signals (abstract skims, deep opens, query refinements, seen-fraction, saved-this-session, revisit)—each normalized to $[0, 1]$. Four design choices answer the four difficulties. Against the *cold start*, the weights ship as a hand-set informative prior (Figure 3) with a strongly negative bias $b = -2.6$, so the agent starts cautious and opinionated rather than blank. Against *censored feedback*, we learn from delayed labels: after a fire the agent waits, applying a positive stochastic-gradient step (learning rate 0.05) if the user opens an item within a 45-second window and a negative step if the window lapses or the user undoes—turning the bandit outcome into a usable gradient. Against *non-stationarity and noise*, weights are clamped to $[-6, 6]$ and a shorter learned vector is pad-migrated (never discarded) when the feature schema grows. Against the *cost budget*, logistic regression is a dot product—it predicts $P(\text{act})$ in microseconds every tick and fires when $P(\text{act}) \geq 0.6$. Crucially, “the model says act” is not “act”: a separate **rate-limit gate**—at most two offers per session, a 90-second cooldown, and a ten-minute suppression after a decline—sits between the classifier and any visible action, so a miscalibrated probability can never become a barrage. This separation of a soft probabilistic estimate from a hard safety budget is a core part of our timing solution, not an implementation detail.

Status & evaluation. The classifier, its online update, weight clamping, and the rate-limit gate are all *built and unit-tested* (learning direction, bound enforcement, and budget/cooldown/suppression are covered; Section 5.2); what is not yet established is whether the learned timing matches real journeys. The timing dimension [T] measures this directly: median lag from need-signal to delivery (target < 2 s) and the fraction of hints that expire unseen (target $< 20\%$), read off the intervention

ledger. Offline, the same ledger supports replay regression in the style of PIRA-Bench (Chai et al., 2026); the learning rate, the 0.6 threshold, and the 45-second window are the hand-set knobs M0 shadow-mode is meant to calibrate before any of them is trusted.

3.3 RQ3 — WHAT ORDERING TO DELIVER?

Question and forcing requirement. FR-2 requires need inference whose common case runs locally in under 50 ms, FR-6 requires that taste be learned and remembered per site, and the whole delivery path must satisfy NR-3 (reversible) and NFR-1 (instant when precomputed). The research question: conditioned on the decision to act, produce an *ordering* of the items already on the page that matches the user’s live need—the value($a_t \mid n_t$) side of Equation 1—from a taste model built only from implicit signals. (The separate question of when re-ordering is not enough and the agent must instead *ask* is RQ4–RQ5.)

Why it is hard. (i) *The need is only partially specified.* There is no query and no click on a target; the agent must infer an ordering from a taste model assembled out of implicit, sparse, and noisy engagement signals. (ii) *The content is mixed-language.* The prototype’s sites span Chinese video listings and English research/ML pages, so a single per-site preference representation has to place “crime” and its Chinese equivalent near each other and far from “romance”—a cross-lingual constraint a monolingual encoder cannot meet. (iii) *Cold start recurs here too:* a freshly visited site has no profile, and acting on an empty profile is worse than not acting. (iv) *The common case must be instant and reversible,* with no network round-trip on the hot path.

Formalization. Let the visible items be $\{x_i\}_{i=1}^N$ with embeddings $e_i = f_\theta(x_i) \in \mathbb{R}^{384}$ from the on-device encoder f_θ , and let the per-site preference vector be a signal-weighted centroid of engaged items,

$$c = \frac{\sum_{j \in \mathcal{E}} s_j e_j}{\left\| \sum_{j \in \mathcal{E}} s_j e_j \right\|}, \quad s_j > 0 \text{ for a search/open/long-watch, } s_j < 0 \text{ for a bounce,} \quad (4)$$

where $\mathcal{E}$ is the engagement history and s_j the behavior-derived signal weight. Delivery is the permutation that sorts items by cosine similarity to c ,

$$\pi^* = \arg \max_{\pi \in \mathcal{S}_N} \sum_{i=1}^N \gamma_i \cos(e_{\pi(i)}, c) \iff \text{sort } i \text{ by } \cos(e_i, c), \quad (5)$$

with γ_i a decreasing position weight, *executed only if* the confidence gate $|\mathcal{E}| \geq 3$ holds (otherwise $\pi^* = \text{identity}$: stay silent). The hardness is that c must be estimated from implicit s_j alone, f_θ must be cross-lingual so that $\cos(e_i, c)$ is meaningful across the user’s languages, and the whole of Eq. 5 must evaluate within the render budget with cached e_i .

Related work and the residual gap. Ranking to an implicit, evolving need is the most crowded of our neighborhoods, and the crowd is almost entirely server-side. Agentic recommenders such as A-RAG (Yousefi Maragheh et al., 2025) rank candidates for an inferred need and demonstrate that a taste signal can be assembled from behavior rather than explicit ratings—but they run as cloud services over a catalog, recomputing an ordering the user receives on the next request, not re-ordering the DOM already in front of them within one frame and with one-click undo. In the terms of Eq. 5, they solve the same general object— $\pi^* = \arg \max_{\pi} \sum_i \gamma_i s(x_{\pi(i)}, \text{need})$, an ordering that maximizes a need-conditioned score—and our formulation is the specialization $s(x_i, \text{need}) = \cos(e_i, c)$, a single on-device cosine against a preference centroid. The general ranking objective is theirs; what we specialize is the *scoring function* (to a cached embedding dot product) and the *deployment* (in-place, reversible, sub-render-budget). Long-term intent memory (PASK, Xie et al., 2026) and recurring-need detection (H-MAPS, Nishikawa & Kato, 2026) show that persistent per-user preference improves proactive delivery and motivate our per-site centroid, yet both keep the profile server-side and neither faces the mixed-language constraint our sites impose. There is, however, a dedicated *on-device recommendation* literature that shares our deployment surface and is the closest general form of the RQ3 constraint: EdgeRec (Gong et al., 2020) performs context-aware re-ranking of already-served items on the device in Mobile Taobao, and a line of work on compressing recommender embeddings to fit device budgets—elastic embeddings (Chen et al., 2021) and the

broader survey of the area (Yin et al., 2024)—establishes that on-device ranking is feasible under tight memory and latency. Our RQ3 is a specialization of exactly this object with three tightenings the on-device recsys line does not jointly take: the profile is built from *purely implicit* behavioral signal (no ratings, no labels), the representation must be *cross-lingual* so a single centroid spans the user’s languages, and the re-rank must be *reversible in place* inside a Manifest V3 content script rather than a native app with a cloud-synced item store. The cross-lingual half of the problem is answered by multilingual sentence encoders (Reimers & Gurevych, 2019), which give us a single vector space where Chinese and English items of the same theme sit together—but that literature is about representation quality in the abstract, not about running a quantized encoder single-threaded inside a Manifest V3 offscreen document under a <150 ms render budget. The residual gap for RQ3 is thus not “can we model taste” (the field shows we can) but “can we do it as an *on-device, instant, reversible, in-place* re-rank from purely implicit signal, in a representation that spans the user’s languages”—the conjunction of *deployment surface* and *cross-lingual implicit modeling* that the server-side recommender literature never has to confront. What stays hard is cold start under this constraint: with no explicit ratings and a three-signal floor before the agent may act, a freshly visited site is mute by construction, and the feasibility budget (Section 4.6) forbids the obvious fallback of asking a cloud model to rank every page.

Our current solution. We separate *what to deliver* from *when to act* into two modules with different costs and learning signals—the central architectural claim of the research component (Figure 2), mirroring the split between interruptibility models (Mehrotra & Musolesi, 2017; Zheng et al., 2025) and need inference. The local tier answers the common case with a per-site **preference centroid**: a running mean of the 384-dimensional multilingual sentence embeddings of engaged items, blended toward positive signals (a search query $+0.8$, an open $+0.3$, a long watch) and away from negatives (a bounce), at rate $\alpha = 0.35$. A re-rank fires only once the profile has at least three signals (below that the agent stays silent rather than rank on noise), scores each visible card by cosine similarity to the centroid, and re-orders in place; a stored vector of mismatched dimension is discarded rather than blended, so a model change never corrupts taste. The multilingual encoder answers difficulty (ii) directly—we verified a Chinese “crime” query embeds near an English “crime” item (cosine 0.54) and far from “romance” (cosine 0.11).

Status & evaluation. The centroid, confidence gate, dimension-safety, and the in-place re-rank are *built and unit-tested*, and the cross-lingual property is empirically checked; what remains is whether the resulting order matches need in practice. The act-value dimension [V] measures this: hint-engagement rate (target ≥ 25 –30%) and R2 usefulness (explicit $\uparrow$ or dwell proxy, target $\geq 50\%$ of engagements), both from the ledger.

3.4 RQ4 — WHETHER TO ASK A QUESTION?

Question and forcing requirement. Some needs cannot be served by re-ordering what is already present—the agent must *elicit*. But a question demands active attention, so it is among the costliest rungs on the ladder, and FR-5.2 requires it to pass a usefulness gate before it may appear: expected information gain $\times$ value of the downstream action must exceed the interruption cost, or the agent stays silent. Open question Q-6 goes further and asks whether R4 is worth building at all, or whether ambient and additive rungs capture most of the value. The research question is therefore: decide *whether* a clarifying question clears the bar, before committing the user’s attention to it.

Why it is hard. The gate must estimate the value of information it does not yet have. Expected information gain depends on how ambiguous the need actually is (hard to know without asking) and on how much a resolved answer would change the delivered order (knowable only counterfactually); a question that merely confirms what the agent would have done anyway is pure interruption.

Formalization. Let q be a candidate question with answer A (a random variable under the agent’s current belief), and let $U(\pi)$ be the utility of delivering ordering π . Asking is worthwhile only when the expected utility of the *post-answer* ordering, net of the interruption cost κ_q of a question, beats the best ordering available *without* asking:

$$\underbrace{\mathbb{E}_A[U(\pi_A^*)]}_{\text{ask } q} - \kappa_q > \underbrace{U(\pi^*)}_{\text{re-rank silently}}, \quad (6)$$

A two-tier just-in-time agent: a local loop handles the common case; the cloud is reserved for what only it can do

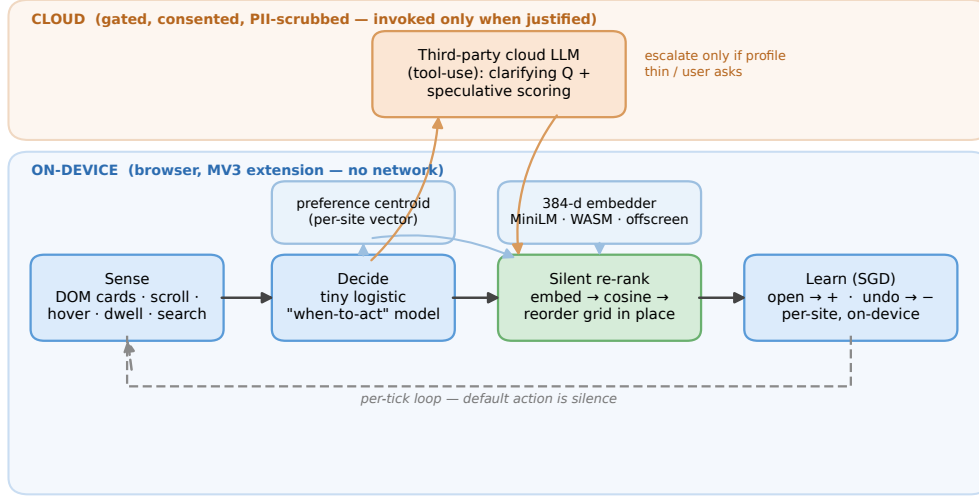


Figure 2: The two-tier decision architecture separating *when to act* (RQ2) from *what to deliver* (RQ3–RQ5). The on-device loop (sense $\rightarrow$ decide $\rightarrow$ silent re-rank $\rightarrow$ learn) runs entirely in the browser with no network access and is the default path; its per-tick default action is silence. The cloud tier (a tool-using LLM) is escalated to only when the local preference profile is thin or the user explicitly asks, and receives only PII-scrubbed snippets.

i.e. the expected *value of information* $\text{VoI}(q) = \mathbb{E}_A[U(\pi_{|A}^*)] - U(\pi^*)$ must exceed κ_q . Two things make Eq. 6 hard to evaluate before acting: the expectation is over an answer the agent does not have, and κ_q is not a fixed turn-cost but the (larger, trust-depleting) cost of an *unsolicited* interruption—so both sides are estimated, and the outside option is the silent re-rank of Eq. 5, not a null action.

Related work and the residual gap. Whether to ask is the best-studied of our five challenges, in a conversational-search literature that nonetheless assumes a different cost structure than ours. The usefulness-estimation line (Sekulić et al., 2024) formalizes exactly our gate—ask only when expected information gain justifies the interruption—and IntentRL (Luo et al., 2026) learns a when-to-clarify policy, while broader proactive-dialogue work (Deng et al., 2023; Lee et al., 2024) studies clarification as a first-class action. The general form here is older and more explicit than the rest: Rao & Daumé III (2018) rank clarifying questions by the *neural expected value of perfect information*—a good question is one whose expected answer is most useful—which is precisely the $\text{VoI}(q)$ quantity of Eq. 6, and Aliannejadi et al. (2019) frame the decision as whether to *return results or ask*, the ask-versus-act choice at the heart of RQ4. This is therefore the tightest general $\rightarrow$ special fit of the five: Sekulić’s usefulness criterion is the value-of-information inequality of Eq. 6, $\text{VoI}(q) > \kappa_q$, stated in general form, and Rao & Daumé give it its canonical EVPI formulation. Our specialization sets two of its terms concretely—the utility $U(\pi)$ is the utility of a delivered *ordering* (Eq. 5), and the outside option against which VoI is measured is the silent re-rank rather than a null turn. But all of it is set in dialogue, where the “cost” of a clarifying question is a conversational turn the user came intending to spend: the human is already talking to the agent and expects to be asked. Our setting inverts that contract—the user came to *read a page*, not to converse, so a question is an unsolicited interruption whose cost is measured in depleted standing trust, and it competes against the option of simply re-ranking silently, which the dialogue setting has no analogue for. The residual gap for RQ4 is therefore estimating usefulness when the *outside option is silence, not a different utterance*, and when a wrong ask costs the agent’s permission to act at all rather than a wasted turn. This is also where the product itself is least certain: open question Q-6 asks whether R4 earns its complexity or whether the ambient and additive rungs capture most of the value, making RQ4 the one challenge whose honest answer may be “defer it.” What stays hard is the counterfactual at the gate’s core—

expected information gain depends on how much a resolved answer would change the delivered order, which is knowable only by imagining the answer we do not yet have.

Our current solution. A clarifying question is escalated only when the local profile is too thin to rank confidently *and* the ambiguity is high-value, and it is budget-capped at ≤ 1 per site per session (FR-3.3). When escalation is warranted, a cloud LLM reads the raw interaction timeline and the actual page titles and returns, through a forced tool call, a behavior reading and a one-sentence need hypothesis alongside the question—so the “is this worth asking” judgment is made against the concrete items on the page, not in the abstract. FR-5.3 closes the loop: an answered question must *visibly* change the order on the current page, so a question that would not change anything fails its own reason to exist.

Status & evaluation. The escalation path and per-site budget are *built*; the usefulness gate is currently a thin-profile heuristic rather than a calibrated info-gain estimate—this is the least mature of the five questions and, per Q-6, may be deferred entirely if M1 shows R1/R2 already capture the value. Evaluation: the gate’s hold-rate (fraction of eligible moments where it correctly stays silent), and the ask→behavior-change rate from FR-5.3 (an answered question that changes nothing is a gate failure), both on the [V]/[R] dimensions.

3.5 RQ5 — HOW TO ASK IT? THE DESIGN OF THE QUESTION ITSELF

Question and forcing requirement. Given that a question clears the gate (RQ4), its *form* is a separate and largely open design problem. FR-5.1 fixes only the outer envelope—one-tap chips, 2–4 options, free text never required—and FR-4.3 requires every rung, questions included, to carry one-line provenance (“why am I seeing this”). Everything inside that envelope is unspecified, and getting it wrong inflates dismissal and poisons the very feedback the learner in RQ2 depends on. The research question: within that envelope, what question design maximizes answer rate and information gained while minimizing the felt cost of the interruption?

Why it is hard. The prototype currently asks a fixed *two-option* multiple-choice question, and that is a convenient default, not a justified optimum. Several design axes are genuinely open. *Cardinality*: two options are the fastest to answer but can force a false dichotomy onto a need that is really three-way or multi-dimensional; FR-5.1 already permits up to four, and the right number likely depends on how cleanly the visible items cluster. *Rationale*: FR-4.3 mandates provenance, but whether a short reasoning line (“asking because your last three opens were all theory papers”) raises trust and answer rate—or instead adds reading cost to an act whose whole point is to be cheap—is untested. *Modality*: a chip is not the only option; a scoped highlight-and-confirm, or a reversible “I’ve reordered assuming X—undo?” framing, turns the same elicitation into a lower-attention act. *Grounding*: a question phrased over the actual titles on the page (“more like *Attention is All You Need*, or *BERT*?”) may be answered more readily than an abstract axis (“theory or applications?”). Each of these changes both the answer rate and the quality of the resulting signal, and they interact.

Formalization. Given that RQ4 licenses a question, RQ5 chooses its *form* g from a design space $\mathcal{G}$ spanned by cardinality $k \in \{2, 3, 4\}$, a rationale flag $\rho \in \{0, 1\}$, framing m (chip vs. reversible reorder), and grounding b (abstract axis vs. on-page titles). Each form induces an answer probability $\alpha(g)$, an information gain $I(g)$ when answered, an attention cost $\kappa(g)$, and—the coupling term—an expected degradation $\lambda \delta(g)$ of the RQ2 timing feedback when the question is dismissed or mis-answered. The design objective is

$$g^* = \arg \max_{g \in \mathcal{G}} \left[\alpha(g) I(g) - \kappa(g) - \lambda (1 - \alpha(g)) \delta(g) \right]. \quad (7)$$

The final term is what separates RQ5 from a conversational-UI choice: maximizing answer rate $\alpha(g)$ alone (e.g. the easiest binary chip) can raise $\delta(g)$ and corrupt the learner of Eq. 3, so g^* is not the most answerable question but the one with the best *net* effect on the whole system.

Related work and the residual gap. The *form* of a clarifying question has been studied, but again under assumptions our setting breaks. Work on generating and ranking clarifying questions (Lee et al., 2024; Deng et al., 2023) and on simulating users to evaluate proactive assistants (Nathani

et al., 2026) treats question quality largely as relevance or answerability within a dialogue, where longer, reasoned questions are acceptable because the user is already conversing. The relationship to Eq. 7 here is different from the other RQs, and worth stating honestly: the field optimizes the first two terms of our objective, $\alpha(g)I(g) - \kappa(g)$ (answerability times information gain, net of question cost), so our formulation is not a *specialization* of theirs but an *extension*—we add the coupling term $-\lambda(1 - \alpha(g))\delta(g)$ that charges a question for the damage a dismissal does to the RQ2 timing feedback. Prior work has our objective minus one term; the added term is exactly what makes question *form* a system-level research problem rather than a UI-relevance one. Idle-time pre-computation (ProAct, Hu et al., 2026) supplies the mechanism we borrow to hide question-generation latency—speculatively preparing likely payloads during dwell—but says nothing about the question’s cardinality, provenance, or framing. Evaluation harnesses for proactivity are emerging (PIRA-Bench over GUI streams (Chai et al., 2026), PARE’s simulated active users (Nathani et al., 2026), and proactive problem-solving benchmarks (Pasternak et al., 2025)), yet none scores an *in-page, one-tap, glanceable* question whose entire budget is a fraction of a second of a user’s attention, and none measures the second-order cost we care about—that a poorly formed question corrupts the very interruptibility feedback RQ2 learns from. The residual gap for RQ5 is that the question here is a *UI micro-interaction under a sub-second attention budget*, not a dialogue turn, so the open axes—cardinality, whether to show a rationale, chip-versus-reversible-reorder framing, and grounding in the actual on-page titles—have no answer in a literature written for conversational agents. What stays hard is the coupling: the same choice that maximizes immediate answer rate may degrade the quality of the resulting training signal, so RQ5 cannot be optimized on answer rate alone.

Our current solution. We build the simplest point in the space—a two-option chip whose options are generated by the LLM to split the visible items along the most decision-relevant axis, with a behavior-reading and need-hypothesis available as provenance (FR-4.3), and latency hidden by speculatively scoring both candidate answers during the user’s read time so the re-rank after a tap is effectively instant. This is deliberately a *baseline*, not a claimed optimum: it establishes the envelope and the instrumentation, and leaves the design axes above as the primary open research contribution of the delivery side. Throughout, only PII-scrubbed snippets cross the network, the tier is off by default and consent-gated, and blocklisted categories never invoke it (NR-5).

Status & evaluation. Built: the two-option grounded chip with speculative scoring. Not yet done: any comparison across the design axes. The evaluation is an A/B/n study over the ledger—cardinality (2 vs. 3–4 options), rationale (provenance line shown vs. hidden), and framing (chip question vs. reversible reordered-assuming-X)—scored on answer rate, dismissal rate, and the downstream usefulness of the order the answer produces. Because a dismissed or mis-answered question also corrupts the RQ2 learner’s feedback, question design is evaluated not only on its own answer rate but on whether it degrades timing-model learning—a coupling that makes RQ5 a research question rather than a UI detail.

4 SYSTEM AND ENGINEERING COMPONENT

We now describe how the design of Section 3 is realized as a Manifest V3 browser extension built with the WXT framework (Aklinker1, 2024) in TypeScript. The system comprises a content script (sensing, re-ordering, UI), a background service worker (state, the cloud broker, and the embedder proxy), an offscreen document (the WASM embedder), and a local storage layer.

4.1 ON-DEVICE EMBEDDING UNDER MANIFEST V3

Ranking requires turning item text into vectors. Manifest V3 forbids remotely hosted code (Google Chrome Developers, 2024), so the model and its ONNX runtime are *bundled* as extension resources and nothing is fetched at run time. The service worker cannot host the embedder—MV3 blocks the `dynamic import()` the WASM runtime needs—so the model runs in an *offscreen document*, and the service worker proxies embedding requests to it. We use *paraphrase-multilingual-MiniLM-L12-v2*, a 384-dimensional symmetric-similarity sentence encoder (Reimers & Gurevych, 2019), quantized to 8-bit and executed single-threaded (the browser gives the worker no cross-origin isolation, so multi-threaded WASM is unavailable). A multilingual model is a deliberate choice over a language-specific one: it lets a single preference

vector rank mixed Chinese and English content, which the video sites and the research/ML sites respectively demand. We validated the cross-lingual property directly: a Chinese “crime” query embeds close to an English “crime” item (cosine 0.54) and far from an English “romance” item (cosine 0.11).

4.2 THE PREFERENCE CENTROID AND THE SILENT RE-RANK

Each site has a local *preference vector*: a running centroid of the embeddings of items the user engaged with, blended toward positive signals and away from negative ones. Engagement is read from behavior—a search query is a strong positive (+0.8), opening an item a mild one (+0.3), and a departure is scored by watch time (a long watch ≥ 120 s is a strong positive; a bounce < 20 s a negative). The centroid is updated by an exponential blend with rate $\alpha \cdot \text{signal}$ ($\alpha = 0.35$), and a re-rank fires only once the profile has accumulated at least three signals—below that, the agent stays silent rather than act on noise. When it fires, the re-rank embeds every visible card, scores each by cosine similarity to the preference vector, and re-orders the list; a mismatched-dimension stored vector (e.g., from a previous model) is discarded rather than blended, so a model change never corrupts the centroid.

4.3 THE ONLINE “WHEN-TO-INTERACT” CLASSIFIER

The timing module is a logistic-regression classifier over a 16-dimensional journey feature vector (Figure 3). Features summarize the current session: scroll distance, hovers, opens, bounces, search/filter “hunts”, dwell time, time since last open, a no-open-yet flag, dissatisfaction (short last watch), late-night, and six research/discovery signals (abstract skims, deep opens of a PDF/code artifact, query refinements, the fraction of the list already seen, whether something was saved this session, and whether the listing is a revisit). Each feature is normalized to $[0, 1]$. The classifier predicts $P(\text{act})$ in microseconds on every decision tick; it fires a silent re-rank when $P(\text{act}) \geq 0.6$ and a rate-limit gate (at most two offers per session, a 90-second cooldown, and a ten-minute suppression after a decline) permits it.

The classifier’s initial weights (Figure 3) encode a hand-set prior: bounces, hunts, dwell, and no-open-yet push toward acting, while additional opens, deep opens, and a recent save push away from it; the bias is strongly negative ($b = -2.6$) so the default state is not to act until evidence accumulates. From there, the model learns *online and per site*: after each fire, the agent waits, and if the user opens an item within 45 seconds it applies a positive stochastic-gradient step (this state warranted acting); if the window passes with no engagement, or the user undoes, it applies a negative step. Weights are clamped to $[-6, 6]$ to resist runaway from noisy signals, and a shorter learned weight vector is pad-migrated (not discarded) when the feature set grows, so a schema bump does not wipe learning.

4.4 NON-DESTRUCTIVE RE-ORDERING OF THE LIVE DOM

Re-ordering a live page without breaking it is the riskiest operation (non-negotiable requirement NR-6) and the one most carefully engineered. The re-order engine never removes nodes: it resolves each card to its movable “unit”—which may be a single grid cell, or a group of sibling elements that must travel together (an `arXiv <dt>+<dd>` pair, a three-row Hacker News story)—and re-orders units within their shared container using the CSS `order` property plus a physical re-append, so both flex/grid and block/table layouts reorder correctly. Before the first move the original child order is snapshotted, making restore exact and one-click. Event listeners survive because nodes are moved, not recreated; lazy-loaded images on moved cards are nudged to eager loading so a re-rank does not blank thumbnails. A “hide-seen” mode reversibly demotes already-opened items by the same group-aware machinery. All of this is covered by unit tests (Section 5.2).

4.5 THE GATED CLOUD TIER

When the local profile is too thin to rank, or the user explicitly asks, the agent may escalate to a third-party cloud LLM through a tool-use interface (our prototype uses Anthropic’s Claude, but the tier is model-agnostic). The model is given the raw interaction timeline and the actual titles on the page and returns, through a forced tool call, a single multiple-choice clarifying question with exactly

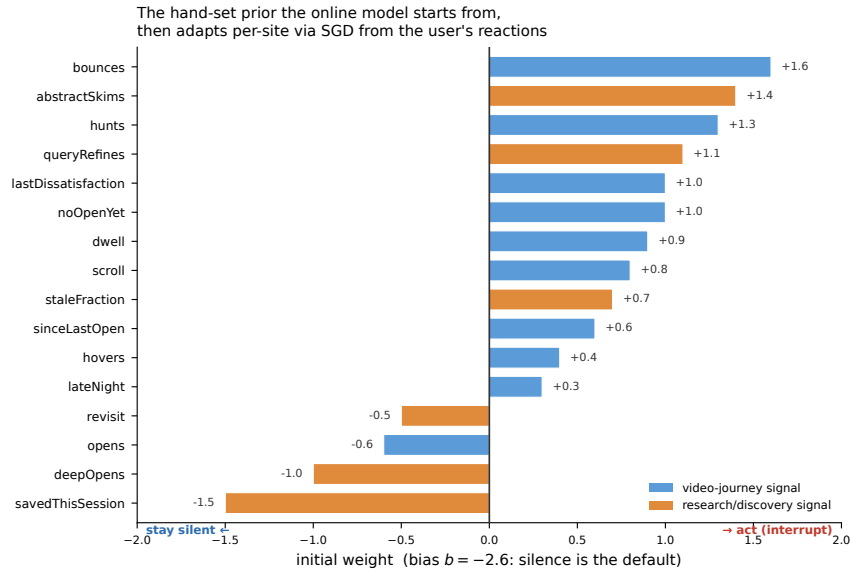


Figure 3: The hand-set prior of the online “when-to-interact” classifier (`src/rerank/policy.ts`). Positive weights push toward interrupting; negative weights toward silence. The strongly negative bias ($b = -2.6$) makes silence the default until journey evidence accumulates. These are only the starting weights: the model adapts them per site via SGD from the user’s reactions.

two options that split the visible items along the most decision-relevant axis, plus a behavior reading (its own positive/neutral/negative classification of the user’s actions) and a one-sentence hypothesis about the current need. A second call scores each item 0–1 given the answer. To hide the two calls’ latency, the agent *speculatively* scores both candidate answers during the user’s read time, so the re-rank after the user taps is effectively instant. Only PII-scrubbed snippets cross the network, the tier is off by default and consent-gated, and blocklisted page categories never invoke it.

4.6 WHY EMBEDDINGS, NOT AN LLM: A LATENCY AND COST FEASIBILITY ARGUMENT

The design choice that most shapes the system is using an on-device *embedding* model for ranking rather than an LLM. This is not a stylistic preference; it is forced by the latency budget (NFR-1: a visible action must render within ~ 150 ms of its breakpoint) and the cost ceiling (NFR-5: a side-project cloud budget). Table 3 makes the comparison concrete for the common case—ranking a listing of $N = 30$ items against the user’s preference—with the two runnable strategies measured directly.

Three consequences follow from these measurements, and the argument is structural rather than tied to the exact figures. (i) *The hot path must be local and non-generative.* The embedding-plus-cosine strategy is the only one that plausibly fits NFR-1, because once an item’s embedding is cached, re-ranking on every scroll or filter change is a set of dot products—which is what would make the “silent, continuous” behavior affordable at all. An LLM in the hot path turns each re-rank into a network- and generation-bound stall. (ii) *Generative cost is real and recurring, so it must be rationed.* Routing every rank through a cloud model incurs a standing per-page cost for work the user never asked for; NFR-5 therefore derives per-page call caps from a fixed budget (FR-8.2, default ≤ 2 LLM calls/page), and the architecture spends those calls only on the elicitation case (RQ4–RQ5) where a generative model does something an embedding cannot. (iii) *On-device LLMs do not yet close the gap.* A browser-built-in small model removes the cost and privacy objections but not the latency one: emitting a full ordering token-by-token is far too slow for a per-scroll action, and current per-prompt input limits do not comfortably hold a 30-item ranking prompt. This is why the embedding model—which produces a fixed-size vector in one forward pass rather than autoregressively generating an ordering—is the right on-device primitive today, and why the LLM

Strategy	Latency (rank 30)	Cost	Functional correctness (format / coverage / no-break)	Verdict (NFR-1 / NFR-5)
On-device embedding (current): 384-d MiniLM q8, WASM, then cosine	~490 ms warm (median of 4 runs, 483–494 ms); cold-load ~1–3 s once	\$0 (fully local, no key)	✓ <i>Correct by construction</i> : one finite cosine score in $[0, 1]$ per item, so coverage is 100%; the score map is built in code, with no model text to malformed. The only no-output path (a stale-dimension vector) is guarded and no-ops the reorder. Cannot break the page.	✓ NFR-5 (zero cost); ✓ NFR-1 (the embedding runs async off the critical path, visible reorder <150 ms; also meets NFR-3). Default.
Cloud LLM (third-party, e.g. Claude Haiku 4.5): score via forced tool call	~3.6 s median (3 real calls; / re-rank ~1.49 k in / ~640 out tokens)	~\$0.0047 (~\$4.70/1k); prefix below the 4096-token cache floor, so no cache discount	✓ <i>Correct by measurement</i> : 11/11 runs fully correct (100% coverage, all IDs mapped, scores in range) across single-chunk, a 45-item cross-chunk merge, and adversarial (injection-shaped) titles. Degrades gracefully: missing scores default to 0 so items stay listed, and a malformed response is caught and falls back to the on-device re-rank (NFR-4).	△ NFR-1: ~3.6 s just over the ≤3 s cold-LLM ceiling—acceptable only behind explicit engagement. △ NFR-5: real cost → budget-capped (FR-8.2), BYO-key, off by default.

Table 3: Feasibility of the two runnable ranking strategies for the common case (rank $N = 30$ short items), measured on a single machine (2026-07-17). The functional-correctness column reports whether each strategy produces a well-formed, complete ranking that never breaks the page. (An on-device browser LLM such as Gemini Nano via Chrome’s Prompt API is a third option in principle, but the Prompt API is not exposed in our headless test runtime, so we do not report figures for it.) *Measurement*: the embedding path is timed as cold embed (per short title, single-threaded WASM) plus cached cosine re-rank; the cloud path is measured over three real forced-tool-call requests (~1.49 k input / ~640 output tokens), priced from usage, with functional correctness checked over 11 runs. Two findings are worth flagging. (i) The cloud path’s ~3.6 s is dominated by generating ~640 output tokens (30 structured scores plus a rationale), not the network; it clears the NFR-1 ceiling only because FR-3.2 already gates it behind explicit engagement, and it must never touch the silent rung. (ii) Prompt caching does not help at this size: the ~1.5 k-token request prefix is below Haiku 4.5’s 4096-token cache minimum, so `cache_creation/cache_read` are both zero and every call pays full uncached input—not a problem here since cost is output-dominated, but it means the `cache_control` marker in the adapter is currently a no-op for these small requests.

is reserved for the rare question rather than the frequent rank. Should on-device generation become fast enough, it is a candidate for the RQ4–RQ5 elicitation tier, not the RQ3 ranking hot path. The measured figures in Table 3 bear this out: the warm embedding re-rank (~490 ms, off the critical path) is the only strategy that fits the hot path, whereas the cloud path (~3.6 s, ~\$0.0047/re-rank) is viable only behind explicit engagement.

The functional-correctness column separates two guarantees that are easy to conflate. The on-device path *cannot* be functionally wrong: its score map is computed in code, so every item always receives exactly one valid score and there is no generated format to malformed—its only failure mode is a stale-dimension profile, which is guarded and degrades to a no-op. The cloud path *could* be wrong, because it generates its output, so we measured it: 11 of 11 runs were clean, including the two paths where format breaks usually hide—the cross-chunk ID remapping (45 items split into 40+5) and adversarial, injection-shaped titles. And the pipeline is defense-in-depth: even a hypothetical malformed response degrades (a missing score defaults to 0 and keeps the item; a caught exception falls

back to the local re-rank) rather than breaking the page, satisfying NFR-4. Both measured strategies are thus functionally sound; the difference is that the on-device path is correct *by construction* while the cloud path is correct *by measurement plus graceful degradation*.

5 IMPLEMENTATION AND ENGINEERING VALIDATION

5.1 SITE COVERAGE AND EXTRACTION

The prototype ships adapters for five sites: one video-streaming site (`iyf.tv`) and four research/ML-discovery sites (`arXiv`, Hugging Face, Hacker News, and Papers with Code). Each adapter is a small declarative object specifying how to find item nodes, extract a clean title and topical tags, compute a stable per-item id, resolve the movable unit, and recognize an opened-item page. Topical tags matter for ranking quality: on `arXiv`, for example, the adapter feeds the embedder the paper’s subject classification (e.g., “Machine Learning (cs.LG)”) rather than a generic text scrape.

5.2 VERIFICATION

The system carries 61 unit test cases across 13 suites (run under Vitest with a happy-dom environment), covering the preference centroid’s confidence gating and dimension-safety, the online classifier’s learning direction and weight bounds, the monitor’s budget/cooldown/suppression gates, the group-aware re-order and exact snapshot/restore, per-site memory isolation, and each site adapter’s extraction and opened-page detection. Beyond units, the end-to-end behavior was verified with Playwright loading the *actual unpacked extension* (not mocks): the bundled on-device embedder produces 384-dimensional vectors in the offscreen document; seeding a preference profile floats matching items to the top of a real grid with no network call; and the full cloud flow—grounded question, behavior reading, speculative scoring—runs end to end.

5.3 ENGINEERING PROBLEMS SOLVED ALONG THE WAY

Several MV3-specific obstacles shaped the build. The embedder was moved to an offscreen document because the service worker cannot dynamically import the WASM runtime. The bundle was slimmed by an order of magnitude (from a ~145 MB multilingual model to the quantized 384-d encoder) to make a side-loadable package; the runtime and model are bundled to satisfy the no-remote-code rule, with a CSP that permits only `wasm-unsafe-eval`. Re-ordering had to target the wrapper cell rather than a nested anchor to avoid detaching links; SPA navigation is tracked by polling the route and crediting per-item dwell across it; and every `chrome.*` call is guarded against the context invalidation that a dev-reload triggers. The site selectors were reverse-engineered by headless-rendering the target pages.

6 DISCUSSION, LIMITATIONS, AND HONEST STATE

What is and is not demonstrated. We are explicit that this is a systems and engineering contribution. The architecture runs, is tested, and behaves as designed; what we do *not* yet report is a user-facing evaluation on our three dimensions—act-value (is an act worth it), timing (does it land at a low-cost moment), and restraint (is silence correctly the default). The prototype in its current form is also, by the project’s own phased plan, an aggressive one: it exercises the R3 (transform) and R4 (question) rungs directly, whereas the disciplined path our design implies is to run a shadow-mode phase first—sensing, inferring, and logging while rendering nothing—until stage-one precision is proven, and only then to enable visible rungs. Skipping that phasing was a deliberate choice to obtain a working spike; the non-negotiable requirements of Section 3.1 are what make it safe to do so, since the most aggressive action is reversible and rate-limited.

Limitations. The unpacked bundle is ~85 MB (model plus WASM), acceptable for side-loading but requiring further quantization for a store build. The online learner is exercised in unit tests but has not been tuned against real journeys at scale—its learning rate, thresholds, and the 45-second learning window are hand-set knobs. The eligibility model is deliberately narrow: the system targets server-rendered, structurally regular pages and stays dormant on complex single-page applications,

infinite feeds, and a hard blocklist of sensitive categories (banking, health, auth, webmail, collaborative editors). Cross-site profiling is by design absent; the agent’s memory is per-site and locally wipeable. Most fundamentally, the current scope is deliberately *within-page*, not whole-site: the agent re-orders only the item nodes already rendered in the live DOM (those the site itself chose to display, plus whatever infinite scroll has loaded so far), and it never fetches, paginates, or crawls to enlarge that set. It is therefore doing *re-ranking*, *not retrieval*—it can improve the ordering of the candidate pool the site presents, but it cannot expand it. The practical consequence is that the reachable set is small and site-determined: on a paginated listing it is a single page (often tens of items), and a better match sitting on page five, or elsewhere on the site, is out of reach until the user navigates there. Broadening the candidate set—prefetching adjacent pages, or re-ranking across a session’s accumulated history—is left to future work, and would reopen the latency, cost, and privacy trade-offs that the within-page design currently sidesteps.

The system is its own benchmark. Because no external benchmark fits an in-browser re-ranking agent (Yang et al., 2025b; Chai et al., 2026), the system is instrumented so that every intervention writes a record—(need class, confidence, rung, timing context, outcome)—to a local ledger. That ledger *is* the evaluation dataset. The phased plan reads directly off it: (M0) a shadow-mode pass to calibrate the confidence thresholds τ and establish stage-one precision on the author’s own browsing; (M1) enable the ambient and additive rungs and measure hint-engagement rate, additive-card usefulness, and need-to-delivery lag against pre-registered targets; (M2) enable the transform rung under the earned-tolerance ladder and hold the undo rate below a fixed canary; (M3) add the clarifying-question rung under a usefulness gate. Replay tooling in the style of PIRA-Bench (Chai et al., 2026) and simulated users in the style of PARE (Nathani et al., 2026) provide offline regression tests over the recorded streams. A counterfactual study—timed find-the-item tasks with the agent on versus off—is the eventual arbiter of whether the system is, as intended, *occasionally decisive and never annoying*.

7 CONCLUSION

We presented Kairosrank, a proactive just-in-time agent that—without being asked—learns a user’s browsing taste and silently re-orders web listings to match it, escalating to a cloud LLM only when a large model is the only thing that will do. Its defining move is proactivity under restraint: it decides on its own initiative when acting is worth more than staying silent, and defaults to silence whenever it is not. Running the common path on-device is a means to that end, not the headline—it buys the efficiency and privacy that let a proactive agent run continuously without a per-tick network cost, and remains one of the open problems (Section 4.6) rather than the contribution itself. The research contribution is a concrete, non-negotiable requirement-governed *approximation* of the JIR decision objective as a browser interaction design—heuristic estimators and a hand-structured action ladder in place of a solved POMDP—with a clean separation between deciding *when* to act and *what* to deliver. The engineering contribution is a working Manifest V3 realization of that design: a bundled on-device embedder, a per-site preference centroid, an online timing classifier that learns from reactions, a non-destructive live-DOM re-order engine, and a latency-hidden gated cloud tier. The system is built, tested, and honest about what remains—chiefly the user-facing evaluation it is precisely instrumented to produce.

REPRODUCIBILITY STATEMENT

All quantitative descriptors in this paper (feature counts, weights, thresholds, blend rates, timing windows, embedding dimension, cross-lingual cosine values, test counts, and site coverage) are read directly from the source of the prototype: the classifier prior from `src/rerank/policy.ts`, the preference logic from `src/embed/profile.ts`, the embedder configuration from `src/embed/embedder.ts`, the timing gates from `src/rerank/monitor.ts`, and the re-order engine from `src/rerank/reorder.ts`. Test counts are from static enumeration of the `*.test.ts` suites.

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